

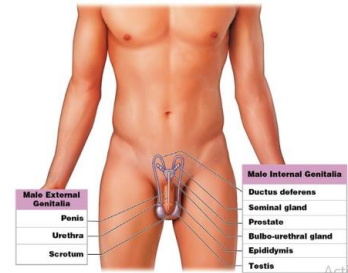
The Uniqueness of the Reproductive System

- **Different structural design** of the organs of the reproductive systems
- Organs of other systems are exactly alike
- Function of the reproductive system is to **perpetuate the species**; i.e., produce new viable offspring
- Function of other systems is to maintain homeostasis of the individual
- Only system **not essential** to sustaining life

Male reproductive structures

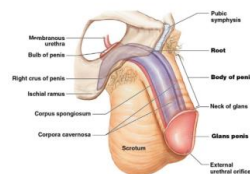
External genitalia

1. Penis
2. Urethra
3. Scrotum



Penis

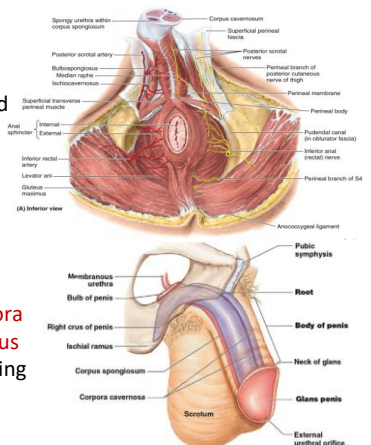
- Is the male **copulatory** organ
- Introduces semen into the female's vagina during sexual intercourse
- Conducts **urine** to the exterior
- consists of:
- ✓ **Root**
- **Fixed** portion that attaches the penis to the **body wall** just inferior to the pubic symphysis



- consists of the crura, bulb, and **ischiocavernosus** and **bulbospongiosus** muscles

✓ Body (or shaft)

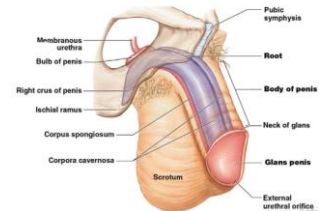
- Tubular, movable portion of the organ
- consists of the **corpora cavernosa**, and **corpus spongiosum** containing the **spongy** urethra



Cont....

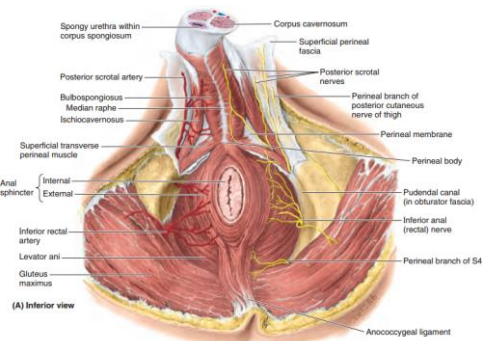
✓ Glans penis (head)

- Expanded **distal end** that surrounds the external **urethral orifice**
- Distally, the corpus **spongiosum** expands to form the conical glans penis

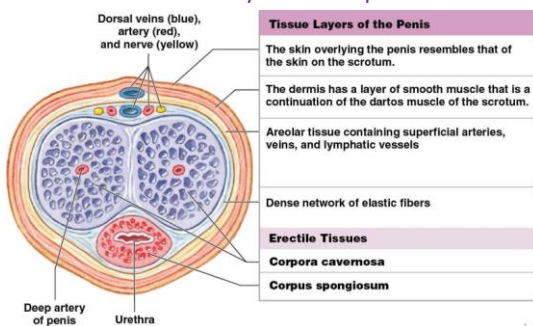


✓ Neck of glans

- Narrow portion between the **shaft** and the **glans**

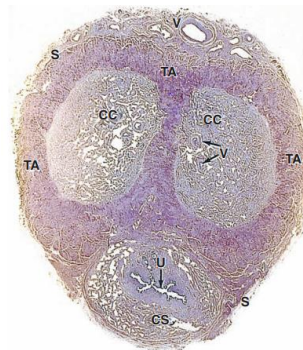


Tissue layers of the penis



- At the neck of the glans, the skin and fascia of the penis are prolonged as a **double** layer of skin, to form the **prepuce (foreskin)**

Cont....

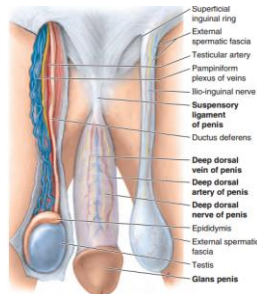


- corpus spongiosum = CS
- urethra = U
- Two corpora cavernosa = CC
- tunica albuginea = TA
- blood vessels = V
- skin = S
- X15. H&E.

Cont....

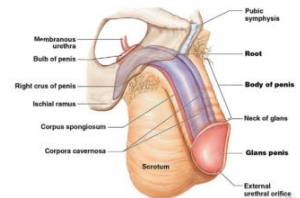
Arterial Supply of Penis

- The penis is supplied mainly by branches of the **internal pudendal** arteries
- branches of the **external pudendal** arteries



Scrotum

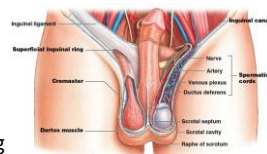
- cutaneous **fibromuscular sac** for the testes and associated structures
- Situated **postero-inferior** to the **penis** and **inferior** to the pubic **symphysis**
- Its external positioning keeps the testes **3°C** lower than core body temperature (needed for sperm production)



Cont....

✓ Scrotal cavities

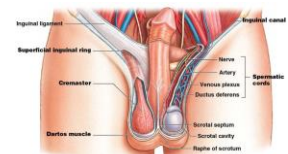
- house the testes
- ✓ Scrotal septum**
- separates **right** and **left** cavities
- Deep** to the scrotal raphe
- prolongation of the **dartos fascia**
- Marked by a raised thickening (**raphe of scrotum**) in the scrotal surface

**✓ Dartos muscle**

- smooth muscle in the **skin** of the scrotum
- Elevates testes and **wrinkles** scrotal surface

✓ Cremaster muscle

- contracts** to pull testes closer to body during sexual **arousal** or when exposed to **cold** temperature
- Extensions of the **internal oblique** muscle of the abdomen



Cont....

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Arterial Supply of Scrotum

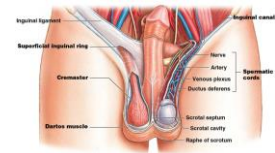
- branches of:
 - femoral artery
 - internal pudendal arteries
 - inferior epigastric arteries

Venous drainage of scrotum

- The *scrotal veins* accompany the arteries

Internal genitalia**1. Testis**

- are the male **gonads**
- paired ovoid reproductive glands that produce **sperms** and **hormones**, primarily testosterone
- suspended in the **scrotum** by the **spermatic cords**



Cont....

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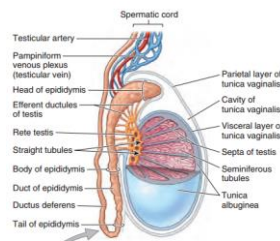
Internal organization of the testes:**✓ Tunica vaginalis**

- Is a closed **peritoneal** sac partially surrounding the testis

- **visceral** layer
- **parietal** layer
- fluid in the cavity allow the testis to move **freely**

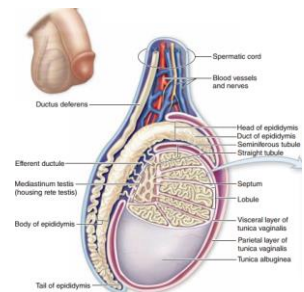
✓ Tunica albuginea

- Tough, fibrous capsule; **covers** testis



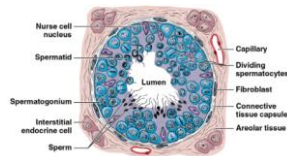
Cont....

- Continuous with **septa** subdividing testes into **lobules**
- ✓ **Seminiferous tubules**
- Coiled tubules within lobules
- Site of **sperm production**
- Myoid cells contract to transport sperm through tubule
- **Spaces between tubules contain:**
 - Areolar tissue
 - Blood vessels



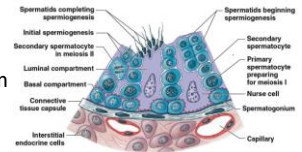
Cont....

- Large interstitial endocrine cells (**Leydig cells**)
- o Produce androgens, such as **testosterone** and
- o **androstenedione** (dominant sex hormones in males)

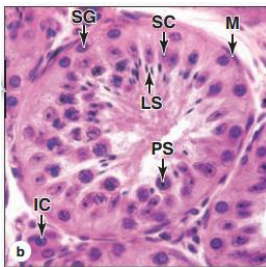


Cells in the wall of the seminiferous tubule

- **Spermatogonia** (stem cells)
- **Spermatocytes** (undergoing meiosis)
- **Spermatids** (undergoing spermiogenesis)
- o Develop into mature sperm
- **Nurse cells (Sertoli cells)**
- o Extend from the tubular capsule to the lumen
- o Located between other types of cells



Cont....



- A seminiferous tubule cross section
- spermatogonia = SG
- Sertoli cells = SC
- primary spermatocytes = PS
- late spermatids = LS near
- interstitial cells = IC
- X400. H&e.

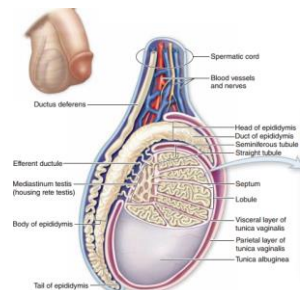
Cont....

✓ Rete testis (*rete*, a net)

- Seminiferous tubules merge into **straight** tubules then rete testes

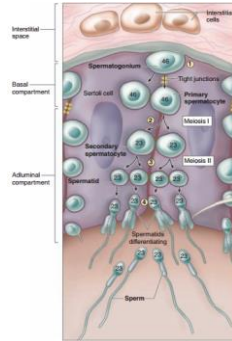
✓ Efferent ductules

- 15–20 tubes connecting the **rete testis** to the **epididymis**



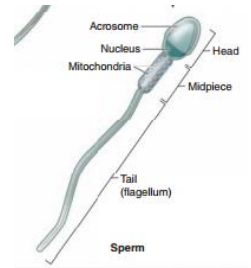
Cont...

- Spermatogenesis (sperm production)



Structure of a sperm

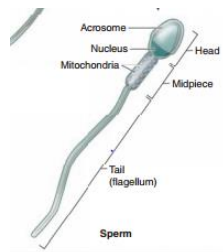
- Specialized to deliver **chromosomes** to female gamete
- Lacks most **organelles** and intracellular structures in order to reduce size and mass
- ✓ **Acrosome**
- Membranous compartment containing **enzymes** essential to **fertilization**
- ✓ **Head**
- Contains **nucleus** with densely packed **chromosomes**



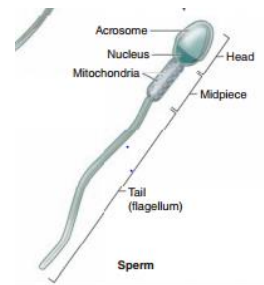
Cont....

Cont....

- ✓ **Middle piece**
- Contains **mitochondria** to provide ATP for tail movement
- ✓ **Tail (flagellum)**
- Whiplike organelle that **moves** the sperm
- Sperm released from testes are physically mature
- But immobile and incapable of fertilizing an oocyte
- **Capacitation** is process enabling sperm to become **motile** and **fully** functional



- Usually occurs in two steps
- Sperm become motile when mixed with **seminal gland** secretions
- Sperm become capable of fertilization when exposed to the **female reproductive** tract

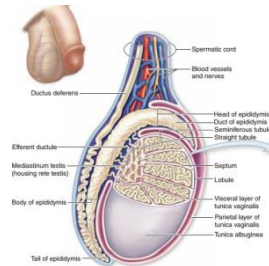


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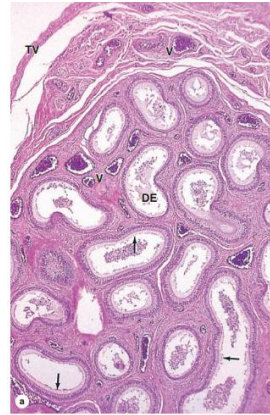
Cont...

Epididymis

- lies in the scrotum along the **superior** and **posterior** sides of each testis
- includes a **head**, a **body**, and a **tail**
- epididymal duct is lined with **pseudostratified columnar**, with small basal cells and tall principal cells bearing long stereocilia



Cont....

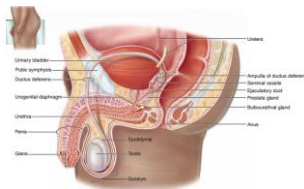


- long, coiled duct of the epididymis = DE
- blood vessels = V
- the tunica vaginalis = TV
- pseudostratified columnar epithelium with long stereocilia = arrows
- . X140. H&e

Cont....

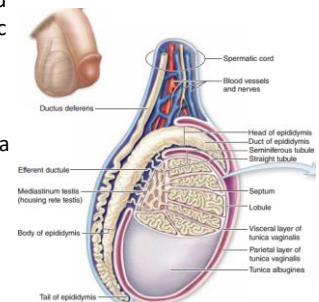
Ductus or Vas Deferens

- a muscular tube approximately **45** cm long
- conveys sperms from the **epididymis** to the **ejaculatory duct** in prostate gland
- Distal end of ductus deferens is enlarged as **ampulla**
- epithelial lining is **pseudostratified columnar**, with fewer stereocilia



Cont....

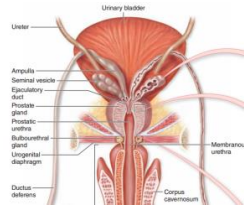
- form the **spermatic cord** with testicular artery and venous plexus, lymphatic vessels, nerves
- Wall of ductus deferens has **smooth** muscle that undergoes peristalsis as a result of sympathetic innervation during ejaculation



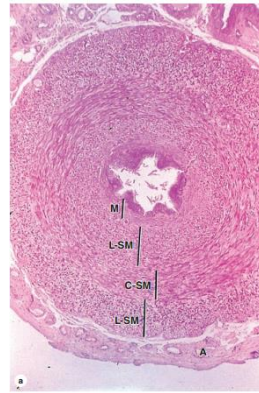
Cont....

Ejaculatory duct

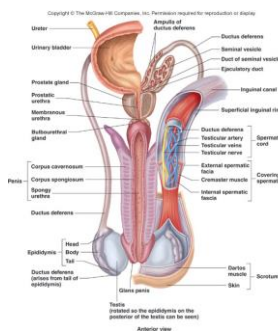
- Joining of ductus deferens and duct of seminal vesicle in the urethra
- Ends at urethra within prostate gland



Cont....



- A cross section of the vas deferens
- mucosa =M
- thick muscularis with inner and outer layers of longitudinal smooth muscle = L-SM and an intervening layer of circular smooth muscle = C-SM
- adventitia =A
- the epithelium is pseudostratified with basal cells and many columnar cells, some with stereocilia
- . X60. H&e.

Urethra

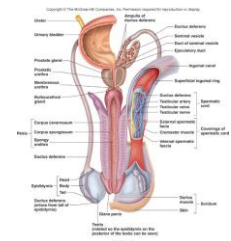
- Extends from urinary bladder to distal end of penis
- Passageway for urine and male reproductive fluids
- Three parts
- I. Prostatic urethra
 - Connected to bladder, passes through prostate
 - 15-30 ducts from prostate empty into prostatic urethra

II. Membranous urethra

- Extends through urogenital diaphragm

III. Spongy or penile urethra

- Passes through the penis
- Several minute mucus-secreting urethral glands empty into spongy urethra



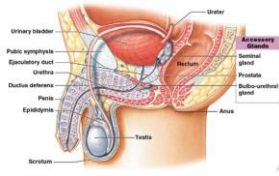
Cont....

accessory glands of the male reproductive tract

- Produce secretions that are **mixed** with **sperm** during ejaculation
- to produce **semen** and that are essential for reproduction

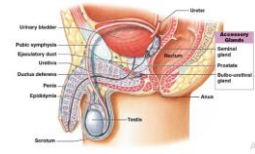
Seminal vesicles

- approximately **5 cm** long
- lies between the **fundus** of the **bladder** and the **rectum**

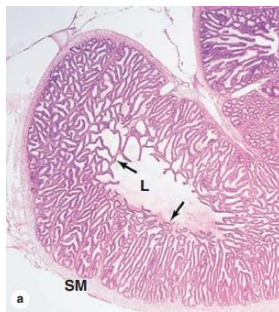


Cont....

- lined with **simple** or **pseudostratified columnar** epithelial cells rich in secretory granules
- Fluid from seminal vesicles makes up about **70%** of the ejaculate and its components include **fructose**, **prostaglandins** and **fibrinogen**



Cont....

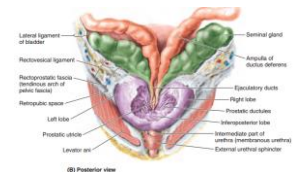


- a coiled duct with a mucosa organized with many thin folds = arrows in the lumen = L
- smooth muscle = SM
- X20. H&E.

Prostate Gland

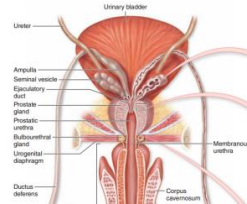
- is the largest accessory gland of the male reproductive system
- surrounds the **prostatic urethra**
- is a collection of 30 to 50 **tubuloacinar** glands embedded in a dense **fibromuscular** stroma
- smooth muscle contracts at ejaculation

Cont....

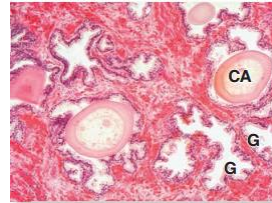


Cont....

- lined by a **simple** or **pseudostratified columnar** epithelium
- concretions called **corpora amylacea** present in the lumens of many prostatic tubuloacinar glands



Cont....



- Individual tubuloacinar glands = G
- corpora amylacea = CA
- X80. H&e.
-

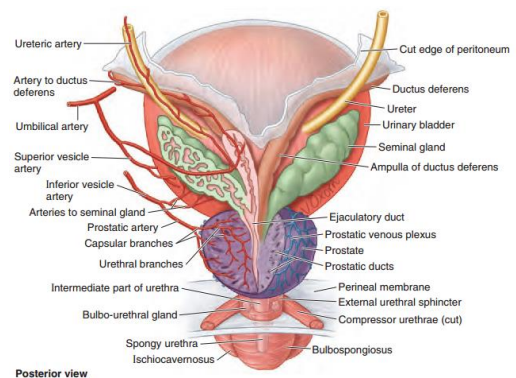
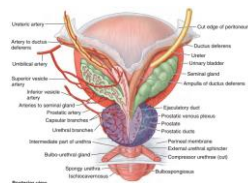
Cont....

Arterial Supply

- prostatic arteries are mainly branches of the internal iliac artery
- inferior vesical arteries*
- Internal pudendal
- middle rectal arteries

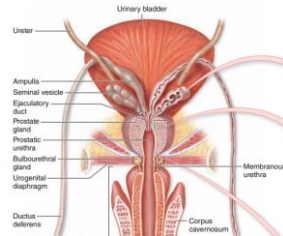
Venous Drainage of Prostate

- prostatic venous plexus
- drains into the internal iliac veins



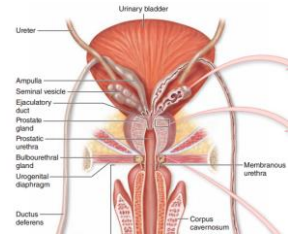
Bulbo-urethral glands (Cowper's glands)

- lie posterolateral to the intermediate part of the **urethra** at the **base** of the **penis**
- empty into the proximal part of the **penile urethra**
- Secrete thick, alkaline mucus



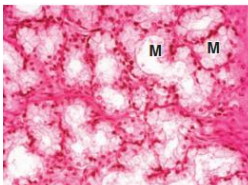
Cont....

- Helps **neutralize** acids in the urethra
- Also lubricates the **tip** of the penis
- Each gland has several lobules with **tubuloacinar** secretory units surrounded by **smooth** muscle cells
- Lined by a **mucus-secreting simple columnar** epithelium

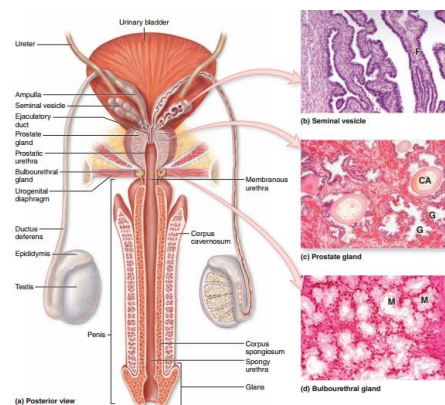


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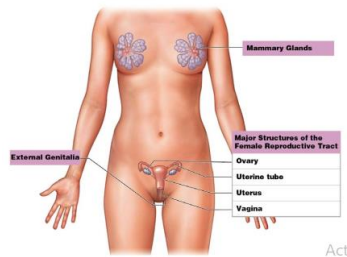


- The small, paired bulbourethral glands consist mainly of mucous acini = M)
- X80. H&E.



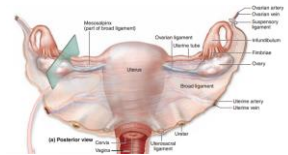
female reproductive system

- Consists of the:
 - paired ovaries
 - oviducts (or uterine tubes)
 - Uterus
 - Vagina
 - external genitalia



Ovaries

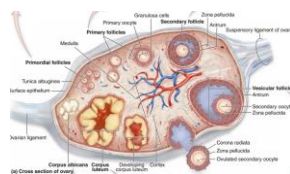
- Paired almond-shaped female gonads
- 3 cm long, 1.5 cm wide, and 1 cm thick
- near the **lateral** wall of the pelvic cavity
- in which the **oocytes** develop



Cont....

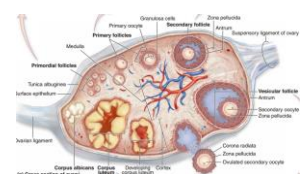
Cont....

- ✓ Have three main functions
 1. Production of female gametes (**oocytes**)
 2. Secretion of female sex hormones (**estrogens** and **progesterone**)
 3. Secretion of **inhibin** (inhibits FSH production in the anterior pituitary gland)



Layers of the ovaries

- ✓ **Germinal epithelium**
 - Layer of **squamous** or **cuboidal** cells covering the ovary
 - Continuous with the visceral peritoneum
- ✓ **Tunica albuginea**
 - Dense connective tissue layer just **deep** to the germinal epithelium



Cont....

Cont...

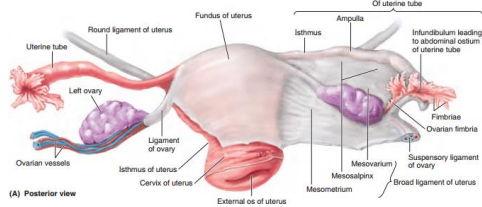
Connective tissues and ligaments stabilize the ovary

✓ Ovarian ligament

- Extends from **uterus** to medial surface of ovary

✓ **Mesovarium**

- Fold of mesentery; supports and stabilizes ovary



Cont...

- ✓ **Suspensory ligament**

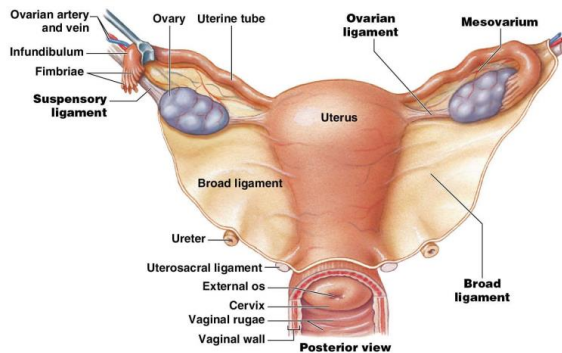
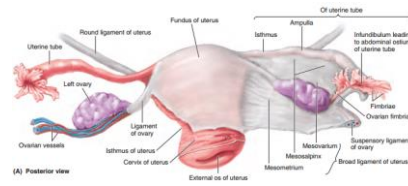
- Extends from lateral **surface** of **ovary** to **pelvic** wall
- ovarian vessels, lymphatics, and nerves

- ovarian vessels, lymphatics, and nerves

- ✓ Broad ligament

- Extensive mesentery enclosing ovaries, uterine tubes, and uterus

- Attaches to sides and floor of **pelvic** cavity



Cont....

Interior of the ovary

✓ **Cortex**

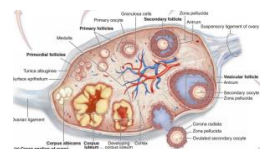
- superficial layer where oocytes are produced

✓ Medulla

- **deep** to the cortex
- contains loose connective tissue and blood vessels entering the organ through the **hilum** from mesenteries suspending the ovary
- **no** distinct border between the cortex and medulla

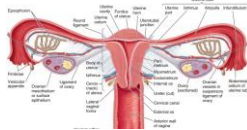
- contains loose connective tissue and blood vessels entering the organ through the **hilum** from mesenteries suspending the ovary

- **no** distinct border between the cortex and medulla



Uterine tubes

- Paired, each measure 10 to 12 cm in length
- extend laterally from the *uterine horns* and open into the *peritoneal* cavity near the ovaries
- provide the usual site of *fertilization*
- lie in a narrow mesentery, the *mesosalpinx*



Cont....

- divisible into four parts, from lateral to medial

1. Infundibulum

- Funnel-like expansion *adjacent* to the ovary
- Has numerous fingerlike projections (*fimbriae*)
- Extend into the *pelvic* cavity
- Drape over the *surface* of the ovary

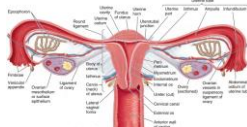


Cont....

- *Inner* surface lined with cilia that beat toward the lumen of the uterine tube

2. Ampulla

- the *widest* and *longest* part of the tube
- begins at the medial end of the infundibulum
- Site where *fertilization* of the oocyte usually occurs



Cont....

4. Uterine part

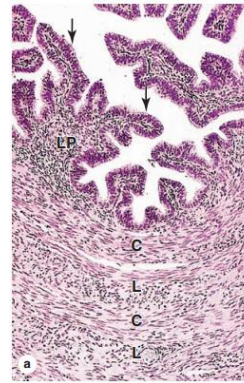
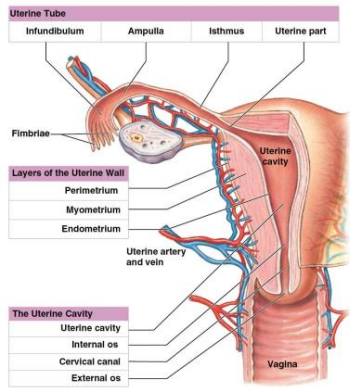
- the short *intramural* segment
- passes through the wall of the uterus and opens via the uterine *ostium* into the uterine cavity at the uterine horn



Cont....

3. Isthmus

- the thick-walled part of the tube, which enters the uterine horn



Cont....

- oviduct mucosa, with folds projecting into the lumen = L
- Simple columnar epithelium = E
- Lamina propria = LP
- . X200. PT

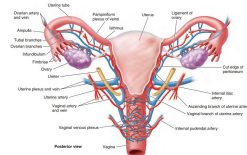
Cont....

Arterial Supply of Ovaries and Uterine Tubes.

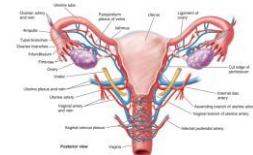
- ovarian arteries
- uterine arteries

Venous drainage

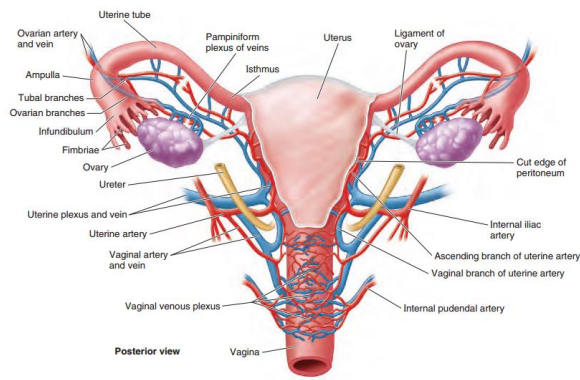
- right ovarian vein ascends to enter the inferior vena cava



- the left ovarian vein drains into the left renal vein
- Tubal veins drain into the ovarian veins and uterine (uterovaginal) venous plexus



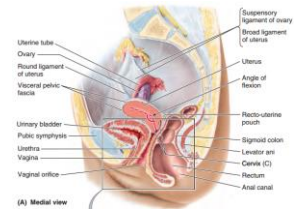
Cont....



Uterus

- is a thick-walled hollow **muscular** organ
- The **embryo** and **fetus** develop in the uterus
- mechanical **protection**, nutritional support, and waste removal
- Contractions in the muscular wall are important in **delivering** the fetus at birth

Cont....



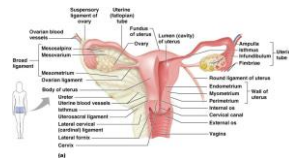
Layers of the uterine wall

I. Perimetrium

- **Outer** surface
- Incomplete serosa continuous with the peritoneal lining

II. Myometrium

- Thick highly **vascularized**, muscular middle layer
- **Smooth muscle** layer provides force for childbirth

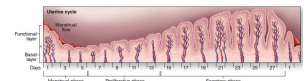


Cont....

III. Endometrium

- Glandular inner lining whose characteristics change with each uterine cycle
- lined by **simple columnar** epithelium
- has two concentric zones
 - ✓ **basal layer**
- adjacent to the **myometrium**

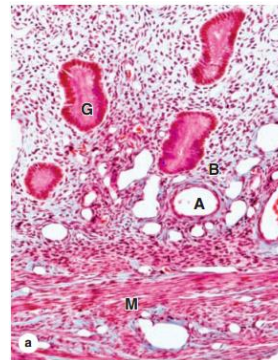
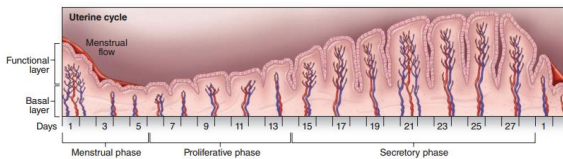
Cont....



Cont...

✓ Superficial functional layer

- undergoes profound changes during the menstrual cycles, but the basal layer remains relatively unchanged



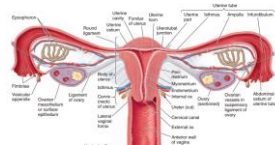
- The basal layer =B
- myometrium =M
- uterine glands =G
- small arteries =A embedded in a distinctive connective tissue stroma with many fibroblasts, ground substance and primarily fine type iii collagen, but no adipocytes. X100. Mallory trichrome

Cont....

Main parts of the uterus

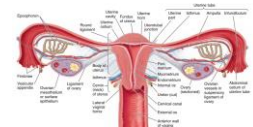
1. Body

- forming the superior two thirds
- includes the fundus of the uterus
- the rounded part that lies superior to the uterine ostia
- Lies between the layers of the broad ligament and is freely movable



Cont....

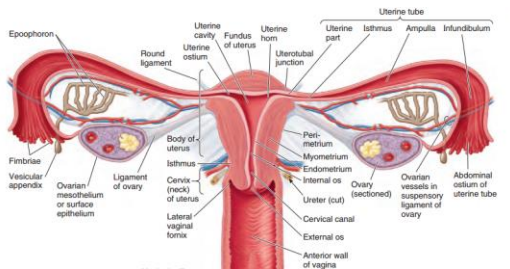
- has two surfaces:
 - vesical (related to the bladder)
 - Intestinal
- The body is demarcated from the cervix by the isthmus of the uterus
- a relatively constricted segment, approximately 1 cm long



2. Cervix

Cont...

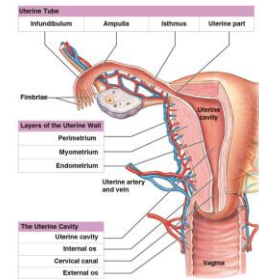
- Inferior portion of the uterus
- Surrounds the cervical canal
- Projects into the vagina



Cont....

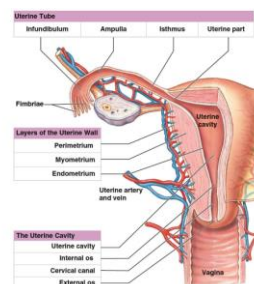
The uterine cavity and cervix

- ✓ Uterine cavity, or uterine lumen
 - Large, superior cavity continuous with isthmus of uterine tube
- ✓ Internal os (os, opening or mouth)
 - Opening connecting the uterine cavity to the cervical canal



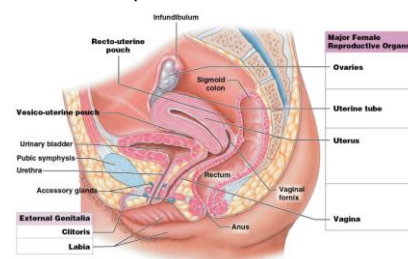
Cont....

- ✓ Cervical canal
 - Constricted passageway at the inferior end of the uterine cavity
 - Begins at internal os; ends at external os
- ✓ External os
 - Curving vaginal opening into the uterus



Cont...

- Vesico-uterine pouch
 - Pocket between uterus and posterior bladder wall
- Recto-uterine pouch
 - Pocket between posterior uterus and anterior colon

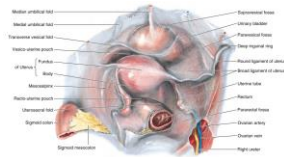


Relations of Uterus

- Peritoneum covers the uterus **anteriorly** and **superiorly**, except for the **cervix**

Anteriorly

- the vesico-uterine pouch and superior surface of the bladder



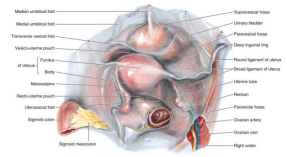
Cont...

Posteriorly

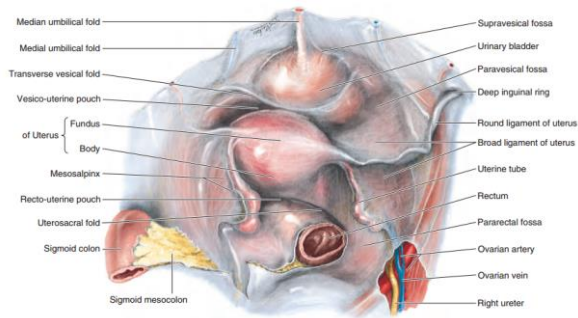
- the recto-uterine pouch containing loops of small intestine and the anterior surface of rectum;

Laterally

- the peritoneal **broad ligament** flanking the uterine body and the fascial **cardinal** ligaments on each side of the cervix and vagina



Cont....

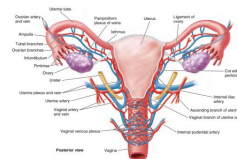


Arterial Supply of Uterus

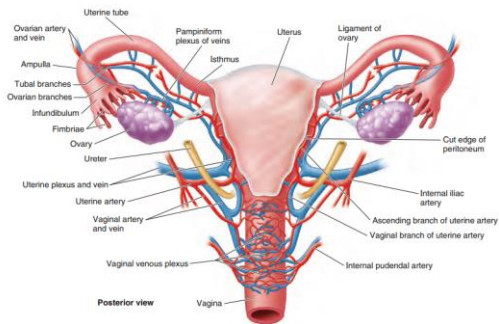
- derives mainly from the **uterine arteries**, with potential collateral supply from the **ovarian arteries**

Venous Drainage of Uterus

- uterine veins

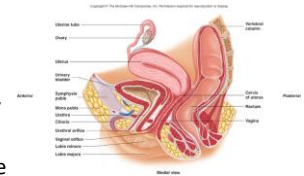


Cont...



Vagina

- Elastic, muscular tube
- Extends from the **cervix** to the **vestibule** (space bordered by the labia minora)
- Typically **7.5–9 cm** long
- Variable diameter (highly distensible)
- Internal passageway is the **vaginal canal**



- The vagina is related:

Anteriorly

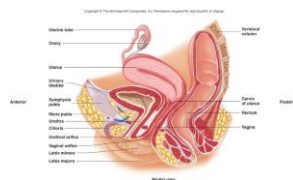
- to the **fundus** of the urinary **bladder** and **urethra**;

Laterally

- to the **levator ani**, visceral **pelvic fascia**, and **ureters**

Posteriorly

- (from inferior to superior) to the **anal canal**, **rectum**, and **recto-uterine pouch**



Cont....

Cont...

• Functions of the vagina

1. Passageway for **menstrual** fluids
2. Receives penis during sexual intercourse and holds spermatozoa prior to their **passage** into the uterus
3. Forms inferior portion of **birth canal**

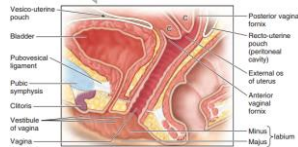
Components of the vagina

Vaginal canal

- Internal **passageway**
- Lined by **nonkeratinized stratified squamous epithelium**

Fornix

- Shallow **recess** in the vagina surrounding the tip of the cervix
- has **anterior**, **posterior**, and **lateral** parts



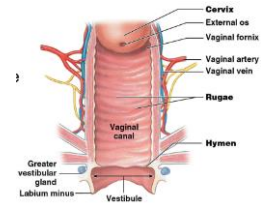
Cont....

Rugae

- **Folds** formed by the vaginal **lining** when relaxed

Hymen

- **Elastic epithelial** fold that usually partially **blocks** entrance to the vagina
- Frequently **absent**
- Stretched or torn during intercourse, tampon use, or heavy physical exercise



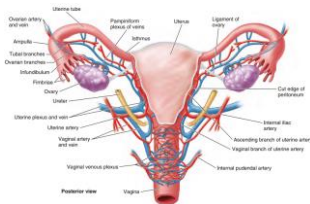
Cont....

Arterial supply and venous drainage of vagina

- Derived from:
 - uterine arteries
 - vaginal and internal pudendal arteries

Vein

- vaginal venous plexuses
- Uterine venous plexus
- Uterovaginal venous plexus



Cont....

Distension of Vagina

- distended, particularly in the region of the **posterior** part of the fornix
- distension allows palpation of the sacral **promontory** during a pelvic examination
- The distension also accommodates the **erect** penis during intercourse



Cont....

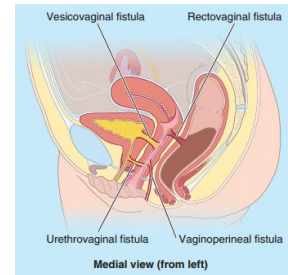
Cont....

- The vagina is especially distended by the **fetus** during **parturition**, particularly in an **AP** direction when the fetus's shoulders are **delivered**
- Lateral distension is limited by the **ischial spines** and the **sacro spinous ligaments**



Vaginal Fistulae

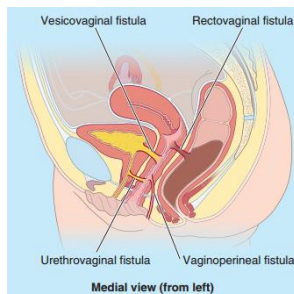
- obstetrical **trauma** during long and **difficult labor** may result in **weaknesses**, **necrosis**, or **tears** in the **vaginal wall** and beyond
- These develop into **abnormal** passages (**fistulas**) between the **vaginal lumen** and the **lumina** of the adjacent **bladder**, **urethra**, and **rectum**, or the **perineum**



Cont....

Cont....

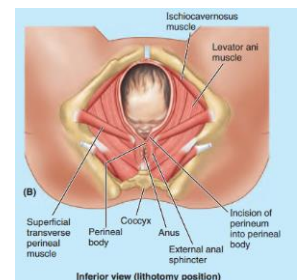
- Urine enters the **vagina** from both **vesicovaginal** and **urethrovaginal** fistulas,
- Fecal matter may be discharged from the vagina when there is a **rectovaginal fistula**



Episiotomy

- surgical incision of the **perineum** and **inferoposterior** vaginal wall
- performed in a large portion of vaginal deliveries
- ✓ **Median episiotomy**
- **perineal body** is the major structure incised

Cont....

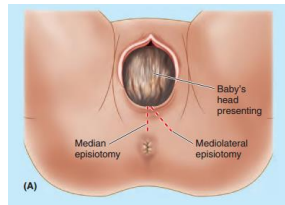


Cont....

- If further tearing occur, it is directed toward the anus, and sphincter damage or **anovaginal** fistulae are potential sequelae

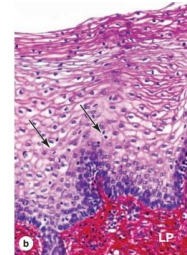
✓ **Mediolateral episiotomies**

- appear to result in a lower incidence of severe **laceration**
- are less likely to be associated with damage to the **anal sphincters** and canal



Cont....

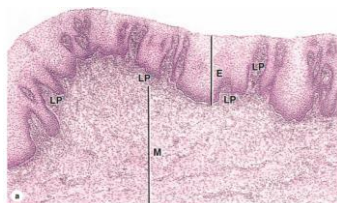
- The wall of the vagina consists of a **mucosa**, a **muscular** layer, and an **adventitia**
- Lined by **stratified squamous nonkeratinized** epithelium
- lacks glands
- Mucus is produced by the **cervical** glands



- lamina propria =LP
- leukocytes = arrows
- X200. PSH

Cont....

- lamina propria =L
- nonkeratinized stratified squamous epithelium =E
- muscular layer = M
- . X60. H&e.

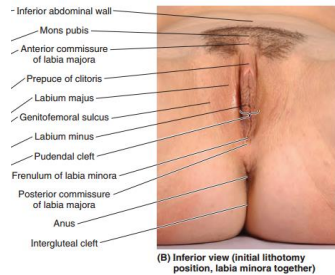


Cont...

- During sexual arousal lubricating mucus provided by glands at the **vaginal** vestibule, including the paired **greater vestibular** glands (of Bartholin),
- which are homologous to the male bulbourethral glands

Female external genitalia

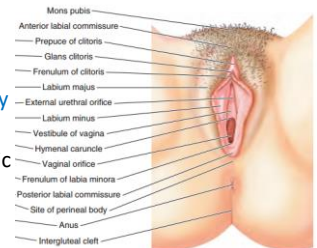
- include the:
 - Mons pubis
 - labia majora
 - labia minora
 - Clitoris
 - bulbs of vestibule
 - greater and lesser vestibular glands



Cont....

Mons Pubis

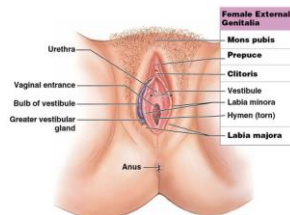
- Fatty eminence **anterior** to the pubic symphysis, pubic tubercles, and superior pubic rami
- Formed by a mass of **fatty** subcutaneous tissue
- covered with coarse pubic **hairs** after puberty



Cont....

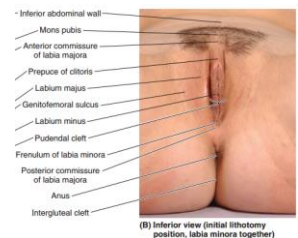
Labia Majora

- are prominent folds of **skin**
- **protect** the clitoris and urethral and vaginal orifices
- passes inferoposteriorly from the **mons pubis** toward the **anus**
- The **internal** aspects of the labia are smooth, pink, and hairless



Cont....

- external aspects of the labia majora in the adult are **covered** with skin containing many **sebaceous** glands, and covered with crisp pubic **hairs**
- lie on the sides of a central depression, the **pudendal cleft**

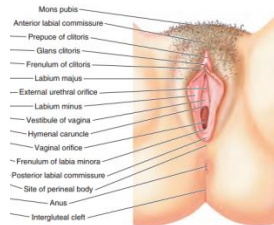


Cont....

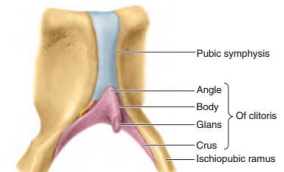
Cont....

Labia Minora

- rounded folds of **fat-free**, **hairless** skin
- enclosed in the **pudendal cleft**
- surround and close over the **vestibule of vagina** into which both the **external** urethral and **vaginal** orifices open

**Clitoris**

- an **erectile** organ
- functions solely as an organ of **sexual** arousal
- consists of a **root** and a small, cylindrical **body**, which are composed of two crura, two corpora cavernosa, and the **glans clitoris**
- The crura attach to the inferior pubic rami and perineal membrane, deep to the labia



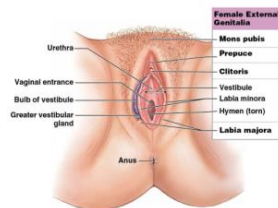
Cont....

Vestibule of Vagina

- space **surrounded** by the labia minora
- orifices of the **urethra** and **vagina** and the ducts of the greater and lesser vestibular glands open

Bulbs of Vestibule

- Masses of erectile tissue on either side of the vaginal **entrance**



Cont....

- Have the same embryonic origin as the corpus spongiosum of the penis

Vestibular Glands

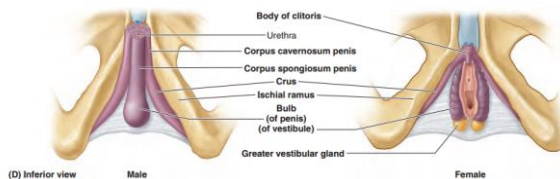
- ✓ **Lesser vestibular glands**
- Secrete onto the vestibular surface, keeping it **moist**
- ✓ **Greater vestibular glands** (Bartholin's glands)
- Activated **during** sexual arousal



Cont....

Cont...

- **Mucous** glands that discharge into the vestibule
- Same embryonic origins as the **bulbo-urethral** glands of males

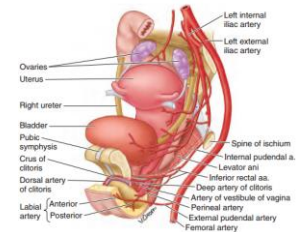


Arterial Supply

- The abundant arterial supply is from the
 - *external pudendal arteries*
 - *internal pudendal arteries*

Venous Drainage

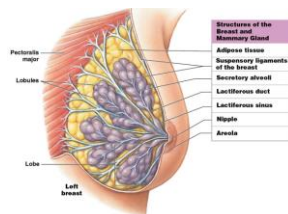
- labial veins are tributaries of the **internal pudendal vein**



Cont....

Breast

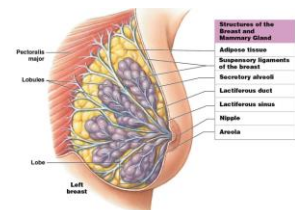
- Each breast contains a mammary gland that **secretes** milk
- Located on the anterior chest, directly over the **pectoralis** major muscle
- Provide nourishment (milk) for developing **infant**



- Milk production (**lactation**) controlled by **hormones** released by the reproductive system and the placenta

Structure of a mammary gland

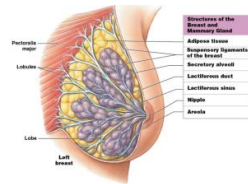
- Embedded in the subcutaneous tissue of the **pectoral fat pad** deep to the skin



Cont....

Suspensory ligaments of the breast

- Bands of **dense** connective tissue
- Surround the **duct** system and form **partitions** between lobes and lobules
- Glandular tissue divided into **lobes**
- Each lobe has several secretory **lobules**
- Each lobule is composed of **secretory alveoli**

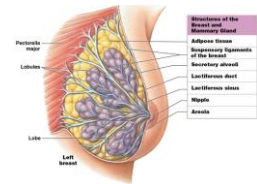


Cont....

- Ducts from the lobules converge into one **lactiferous duct** per lobe
- Each lactiferous duct expands near the nipple to form a **lactiferous sinus**

Nipple

- Conical projection where 15–20 **lactiferous** sinuses open onto the body surface



Cont....

Cont...

- **Areola**
 - Reddish-brown skin around the nipple
 - Grainy texture from sebaceous glands deep to the surface

